BioNGS final presentation

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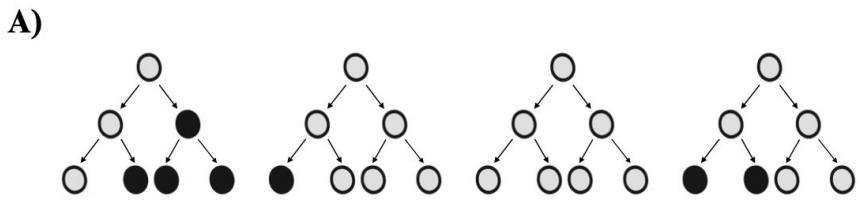
W26 CHEM E 546

Project background: building a NGS analysis pipeline for a novel technique to measure genotypic antibiotic resistance mutation rate



Grow many bacterial populations in parallel

↓

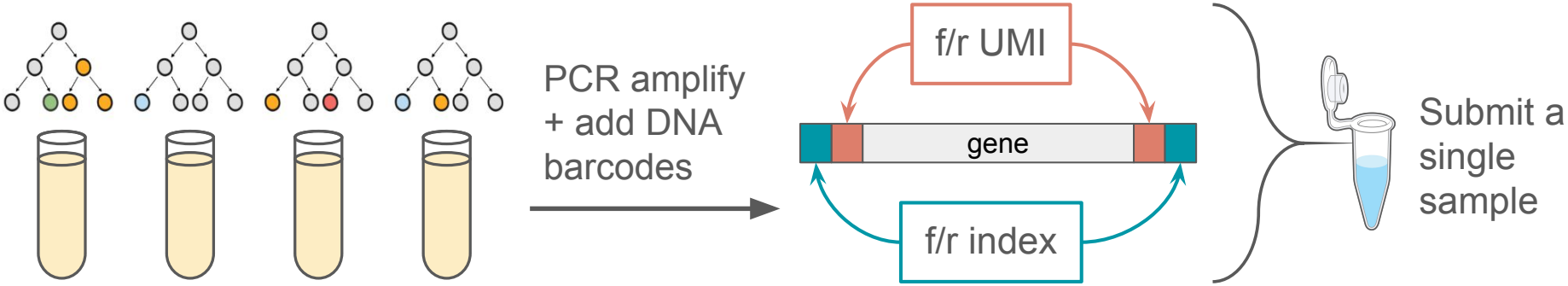
Apply antibiotic to select for mutants

↓

Using the proportion of populations that failed to generate resistance, calculate mutation rate

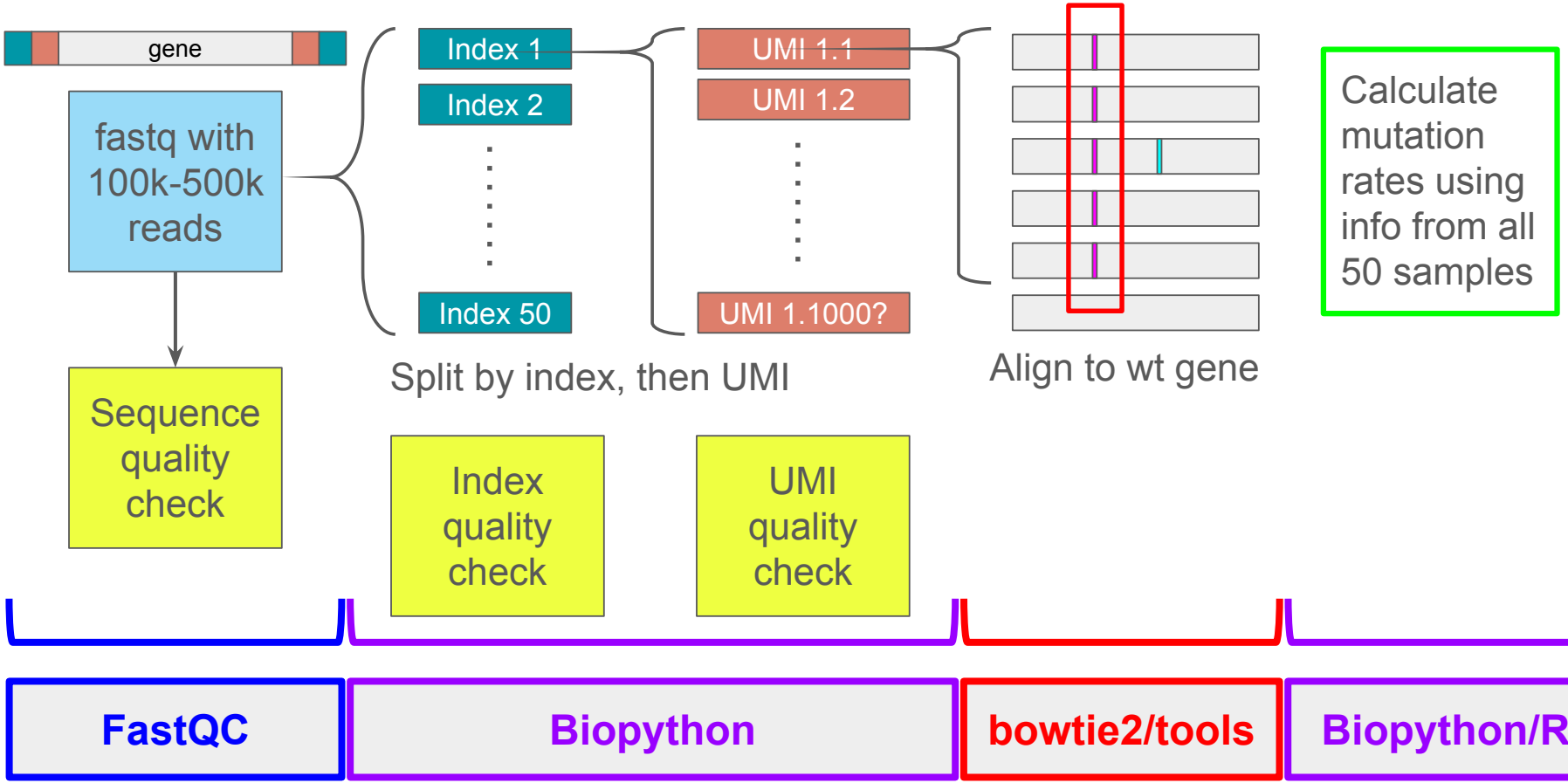
$$\mu = -\ln p_0 \frac{\ln N_t - \ln N_0}{t(N_t - N_0)}$$

Data used: preliminary data from PCR troubleshooting



- What to do if we did not have a full set of data to work with?
  - Preliminary set of data with 2 populations
  - AI generated set of data based on important characteristics of the sequences we expect to get (in a perfect experiment)

Analysis workflow

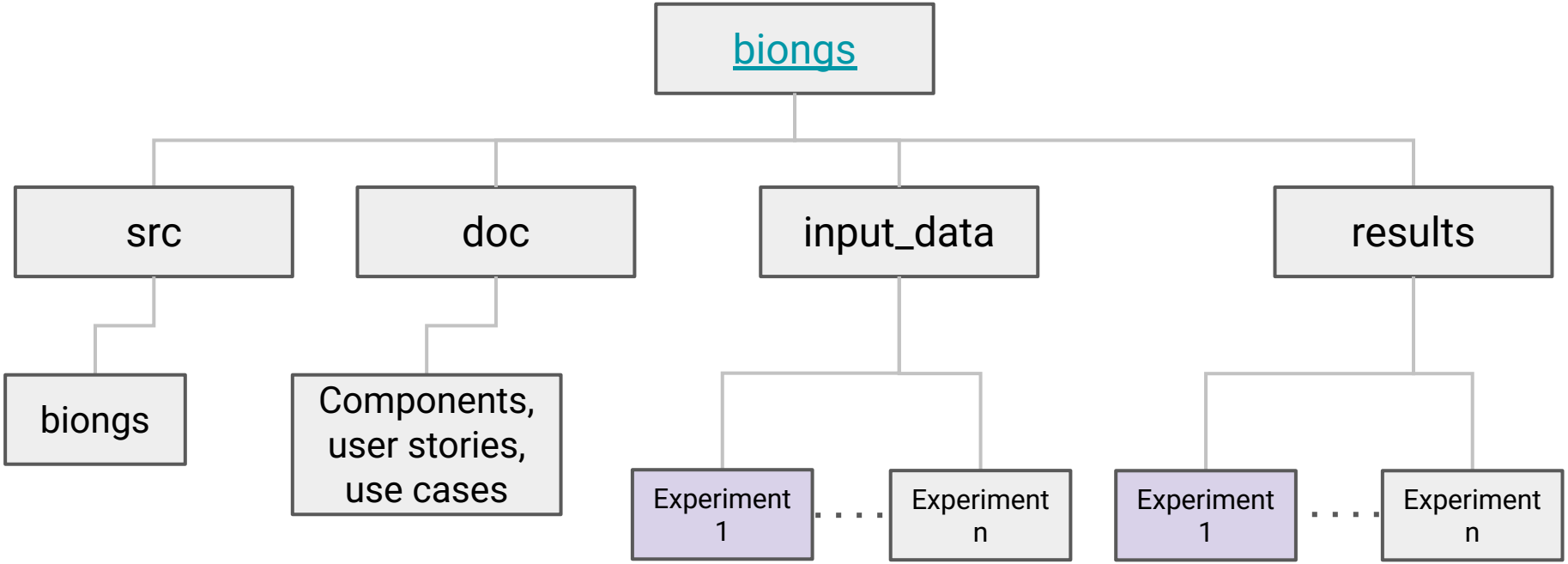


Use cases

- Calculate mutation rate for a given gene
- Troubleshooting the genetic processing protocol done to prepare samples for sequencing
    - Did I amplify the correct gene?
    - Did I submit a clean sample for sequencing?
    - Did my DNA tags get attached to to my gene correctly?
    - Is the number of sequences that map to each tag (UMI, index) sufficient for me to trust my alignment results?

*Can an undergrad with limited understanding of python/bash run the pipeline to analyze their data?

Github project structure



Software demo

- We tested a new genetic processing protocol and just received our sequencing data back! We want to know if our protocol successfully amplified the gene, added tags properly, and added UMI tags at a reasonable frequency

Lessons learned and future directions

Lessons learned:

- Writing a good component spec goes a long way, especially when you are asking AI to write code for you
- How to make a python package, and run tests using github
- How to utilize github from VS *and* the command line

Future directions:

- Continue building out the pipeline (additional components, updating current components)
- Add additional quality checking script (new use cases for genetic processing troubleshooting popping up)